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Parking at University of Tabuk, Saudi Arabia, Problems and Solutions.

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It has been a wonderful & intellectually stimulating experience working on “Parking at University of Tabuk, Saudi Arabia, Problems and Solutions.” which gives us the full knowledge & innovative ideas in the field of Campus Transportation and Parking Lot Design.

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Abstract

This project has applied various concepts related to identifying, assessing, and reducing traffic congestion by organizing parking within the university campus. The aim is to enhance academic performance, safety, environmental impact, accessibility, operational efficiency, and overall satisfaction. By integrating these approaches with data analysis tools, the project seeks to develop more organized and sustainable methods for improving the appearance and functionality of parking areas.

This study serves as a strategic framework, based on collected data, to evaluate the level of congestion and visual distortion in the parking areas of Tabuk University. Consequently, the research provides actionable insights for the university to focus on assessing visual distortion and errors in parking layouts while implementing industrial engineering tools to create well-organized and aesthetically pleasing parking facilities.

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1. Introduction

1.1 General

The term "Traffic Congestion" means a situation in which there are a large number of vehicles moving slowly [1]. It is also said that “congestion is the situation where the introduction of an additional vehicle into a traffic flow increases the journey times of the others” (Thomson and Bull, 2001). The term university campus refers to an institutional space that is designed for use in the education and residence of college students [2], and includes the building and other physical elements found in the associated area [3] also [4] describes the university as a small city because they commonly have their own communities. It's important to look for a solution for our university because the problem has been going on for a long time and a problem like this may have a great impact on different aspects of the life of everyone on the university campus. Through this research, we aim to find solutions to reduce traffic congestion and travel duration by minimizing the number of cars searching for parking, as this process significantly impacts traffic congestion. Some newspaper articles have asserted that during business hours half the cars on downtown streets in large cities are cruising for [14] We can’t overlook how searching for parking can cause traffic congestion as (Zimmerman, 2011) said 30% of urban traffic comes from cars hunting for parking spaces”.

Another reason for the importance of attending to parking is the visual clutter caused by some students parking incorrectly, which sometimes leads to street congestion and an unsightly, unprofessional appearance. Additionally, raising awareness about parking is crucial, especially considering the absence of many basic essentials in the parking facilities. This lack may be due to the university’s oversights rather than the students' actions.

Many of the students now have their own private car and it is normal with how the world is moving forward but it is also normal to witness traffic congestion if we don't do anything about it, The prestige attached to being a car owner is a strong factor in the volume of traffic [1] Most (though not all) experts agree that automobiles will remain central to urban life for the foreseeable future [5].

Developing countries around the world, especially the major urban centers are experiencing fast but imbalanced transport sector development resulting in poor on-road traffic operability due to congestion, and subsequent on and off-road unproductivity attributable to delay [6].

Many studies tried to find solutions for universities traffic through sustainable transportation [3], [7], [8]. Sustainable development “meets the needs of the present without compromising the ability of future generations to meet their own needs.” [9] and at the 2012 Rio+20, they mentioned that sustainable transportation is central to sustainable development, [10].

Creating a successful sustainable transportation network for university campuses is achievable through the formation of a comprehensive and efficient university management structure, [7].

1.2 Problem Statement:

Many local and international universities lack traffic organization, causing disruptions for employees, students, and faculty members due to the absence of effective traffic management. Although numerous studies have been conducted on traffic congestion and ways to mitigate it at other universities, there’s no comprehensive research has yet focused specifically on analyzing traffic congestion at Tabuk University. Poorly distributed working hours lead to traffic

congestion at certain times, increasing pressure on roads. At Tabuk University we suffer from road congestion on multiple roads may be due of many of the students owning private cars or due the type of roads near the buildings do not meet with the capacity of the estimated number of cars for the number of students.

This congestion disrupts the daily routines of students, faculty, and staff, making it necessary for universities to find effective solutions.

A problem like traffic congestion may cause significant dissatisfaction leading to stress, reduced productivity, safety risks, and environmental concerns. Many universities are addressing this issue through sustainable transportation solutions.

1.2.1 Visual distortion

One of the problems resulting from the lack of parking spaces or overcrowding is **visual clutter**. The visual clutter at the University of Tabuk also stems from the absence of curbs and the lack of proper planning in the parking areas. This leads to very poor behaviors that contribute to an unorganized and unattractive appearance, which is entirely unbefitting for a university.

Figures 1. Some pictures contains visual distortion.

1.2.2 The 8:00 AM Congestion

The 8:00 AM congestion is a noticeable issue, characterized by significant overcrowding during this time. This results in disorganized student parking and a high level of chaos in the parking areas.

1.2.3 Staff and Faculty Members

Issues in the parking areas designated for staff and faculty members are evident, with many violations occurring in these spaces. This can lead to frustration and discomfort for employees when they are unable to find an available parking spot, forcing them to resort to violations or improper parking.

Figures 2. Wrong Parking in the staff and faculty members areas.

1.3 Objectives

Our primary objective is to ensure that the University of Tabuk offers a healthy and conducive environment for all its members. This study aims to minimize traffic congestion at the University of Tabuk by collecting data on traffic congestion at the university campus so that we can analyze the patterns to indicate the severity of the congestion and when it frequently occurs by leveraging data analysis and the tools and techniques acquired through our industrial engineering curriculum, solutions such as peak-hour restrictions to limit entry through certain gates, designated loading zones for rideshare pickups and drop-offs to prevent congestion in parking areas, flexible class scheduling can help also in minimizing peak-hour congestion, traffic calming measures by installing speed bumps, traffic islands to slow down traffic and reduce the likelihood of accident or congestion.

Analyzing the parking lots at the university campus is a fundamental component of traffic congestion analysis it can provide valuable insight into capacity, utilization, accessibility, and opportunities for improvement. Providing solutions for the parking lots such as allowing students and staff to park on campus on designated days only based on their classes or work schedule, and enforce of parking regulations consistently to deter illegal parking and encourage compliance with designated parking areas. Drawing parking lot lines to optimize capacity, easy access can contribute to minimizing traffic congestion, improving the efficiency of transportation systems, and prevent students from illegal parking at the campus.

By applying the tools of data analysis, we seek to enhance the transportation services available to students, faculty, and staff. We believe that improving these services will positively impact the academic experience, enabling students to excel in their studies at the university.

2 Literature Review

2.1 Definition of traffic congestion

According to Calderdale Council (2015), traffic congestion is an inherently difficult concept to define as it has both physical and relative dimensions. Traffic congestion can be defined in various ways depending on the context, parameters, and metrics used by different studies and institutions.
The definition of traffic congestion can be categorized into demand-related, delay-related, and cost-related. Demand can represent the cause and on the other hand, the delay and the cost represent the effect.
However, because the exact parameters and thresholds for what constitutes "congestion" can vary, there is no single, universally agreed-upon definition used across all studies and contexts.

TABLE 1. The three dimensions of defining traffic congestion. [11]

Dimension	Author	Definition
Demand-related	Vaziri (2002) [12]	Congestion occurs when traffic demand approaches and exceeds highway capacity
Delay-related	Falcocchio and Levinson (2015) [13]	Congestion in transportation occurs when the occupancy of spaces (roadways, sidewalks, transit lines and terminals) by vehicles or people reaches unacceptable levels of discomfort and delay
Cost-related	Naudé and Tsolakis (2005) [14]	Congestion may be regarded as the point at which an additional road user joins the traffic flow and affects the marginal cost in such a way that the marginal social cost of road use exceeds the marginal private cost of road use at the 'optimal' level of congestion

2.2 Parking Search Caused Congestion:

Several studies have discussed the impact of searching for parking on traffic congestion, and some have proposed innovative solutions. For example, [27] the American University of Beirut examined the experiences of five American universities and concluded that university transport services do not necessarily rely on the existence of an organized public transport system. The study emphasized the importance of designing a campus transport service tailored to the needs of the target population, highlighting that reducing the number of parking spaces or imposing fees on them can significantly decrease the incentive to drive, thereby alleviating traffic congestion.

2.3 Previous Solutions and Studies

Many research studies have attempted to provide useful solutions for traffic congestion problems at universities, and some have focused on solutions based on sustainable transportation.

King Abdulaziz University [17] several solutions were proposed. **First**, coordinating class schedules to distribute them evenly across different days of the week can help reduce traffic density during peak hours. **Second,** expanding high-density gates and adjusting the gate edges can improve vehicle movement. **Additionally**, opening new lanes for right-turning vehicles can alleviate congestion and facilitate smoother entry**. Lastly**, establishing a traffic monitoring room equipped with modern devices will enable the monitoring of traffic at gates, roads, and parking areas, allowing for appropriate interventions to resolve traffic crises.

Kent State University [18] several measures are proposed. Increasing road capacity by building new roads and expanding existing ones, though expensive and time-consuming, effectively improves traffic flow. Utilizing traffic modeling software to analyze and predict congestion areas allows for informed decision-making. Additionally, improving intersection management by adding service roads can further enhance traffic flow.

2.4 Study of Parking Problem on University Campus

Beijing University of Aeronautics and Astronautics [20], This research paper discusses the parking problem on university campuses, using the Beijing University of Aeronautics and Astronautics as a case study. The increase in the number of university employees living off-campus and owning private cars has led to a shortage of parking capacity on campus. On-street parking has become common due to the shortage of parking supply, Leading to traffic congestion and safety concerns. The study surveyed and analyzed the inflow and outflow of vehicles, the location and use of parking lots, and the drivers' parking behavior. The average parking duration and use turnovers of parking lots were computed from the survey data. The research highlights the need for better management of parking on university campuses and suggests solutions for improving the campus parking system. The research suggests implementing better management strategies for parking on university campuses, such as increasing the number of parking spaces, regulating on-street parking, and implementing parking fees and permits to encourage turnover of parking spaces.

2.5 Parking fees

An average American household travels annually about 20,000 miles on roads and spends about 15% of its income on road transportation.1 In 2011, congestion in the US caused around 5.5 billion hours of travel delay and 2.9 billion gallons of extra fuel consumption with a total cost of $121 billion [21]. [22], is that the technology needed to charge for parking is much simpler than that needed to charge for driving in congested traffic.

2.6 Sustainability

Kent State University [19] This research paper examines the level of sustainable transportation, specifically walking and bicycling, on a large university campus in the US Midwest. The paper measures the current level of walking and bicycling around the campus, surveys students about their transportation habits and attitudes, and inventories the campus infrastructure. The findings show that there is a low level of sustainable transportation among students, particularly bicycling. Reasons for this include time and convenience factors, as well as limited bicycle access. The paper suggests that existing infrastructure discourages sustainable transportation activity on campus. The study has implications for other campus communities and suggests that understanding the impediments to walking and bicycling can help universities design more attractive and useful facilities.

King Saud University [8] with some focusing on sustainable transportation. Proposed solutions include improving campus signage to provide clear directional signs, aiding students, staff, and visitors in reaching their destinations easily and safely, thereby reducing congestion and enhancing the effective use of roads and pathways. Additionally, adopting sustainable policies to reduce the demand for private car trips, implementing car-sharing programs, and encouraging the use of alternative transportation methods are crucial. Encouraging environmentally friendly transportation by increasing reliance on public transportation, providing shuttle bus services within the campus and between residential areas and the university, and promoting walking through the creation of green spaces and recreational areas to enhance the walking experience are also important measures.

2.7 Tree Planting

Soil Volume and Planting Dimensions

Studies by Penn State University emphasize the critical role of soil volume in supporting tree health. According to their recommendations, planting pits for trees in sidewalks and roads should measure no less than 1.5 meters by 1.5 meters, with a depth of **0.9 meters. They further stress that "bigger is better" to accommodate the growth of larger trees and ensure their long-term health in construction-damaged soils [24].

Spacing and Tree Placement

Urban forestry experts also underline the importance of adequate spacing between trees. Recommendations from Utah State University suggest a distance of 5 to 7 meters between trees planted in sidewalks with a width of 3 meters. This spacing allows sufficient room for root expansion and minimizes the risk of interference with surrounding infrastructure, such as pavements and utilities [25].

Parking Lot Layout Considerations

The layout and dimensions of parking spaces play a pivotal role in integrating tree planting effectively. According to technical guidelines on car park design, the standard dimensions of :parking spaces vary geographically

.European Union and Switzerland: 5 m long by 2.50 m wide -

United Kingdom: 4.80 m long by 2.40 m wide.

Canada: 5.20 m long by 2.60 m wide.

United States: 18 ft by 96 inches (5.48 m by 2.44 m).

These variations necessitate tailoring planting designs to accommodate regional standards. For example, wider parking spaces in Canada allow for more flexibility in tree placement, while narrower spaces in the UK require more strategic planning to prevent interference with vehicle operations [26].

3. Methodology

Implementation of data analysis and industrial engineering tools can facilitate a deeper understanding of congestion dynamics and inform evidence-based decision-making. By optimizing traffic flow using industrial engineering techniques, universities can improve transportation efficiency, reduce congestion-related stress for the community, and enhance overall campus accessibility and sustainability.

We collected data on student attendance in lecture halls after requesting it from the university to ensure the accuracy of the number of students present at different times throughout the week. As for the number of parking spaces, some were calculated manually, such as those for employees and faculty members. For student parking spaces, they were estimated by calculating the parking area while taking into account curbing and landscaping. Several criteria were used to ensure a sustainable and aesthetically pleasing design suitable for the university.

Vehicle Lanes:
The width of vehicle lanes typically ranges from **6 to 9 meters**, depending on the direction and flow of traffic.

Pedestrian Sidewalks:
Sidewalks for pedestrians are generally allocated a width of **2 to 3 meters**.

Parking Space Dimensions:
A standard parking space measures approximately **5 to 5.5 meters in length** and **2.5 meters in width**, resulting in an area of **12.5 square meters per parking spot**.

Example Calculation:
Given a total area of **1,000 square meters**, the following allocations are made:

300 square meters for vehicle lanes.

50 square meters for pedestrian sidewalks.

This leaves **650 square meters** available for parking spaces.

To determine the number of parking spaces:

$$\text{Number of parking spaces} = \frac{\text{Total parking area}}{\text{Area per parking spot}} = \frac{650}{12.5} = 52 \text{ spaces}$$

This calculation demonstrates how land allocation impacts the number of parking spaces in a given area. [29] [30]

We used the R program for analysis due to the large volume of student attendance data, which was then exported to Excel and Power BI for further processing.

The parking areas were divided into six zones, with multiple colleges grouped into one zone due to their proximity and the overlapping parking patterns of students in these areas. As shown in the university map, the distribution of parking spaces is clearly marked, with each zone represented in a different color.

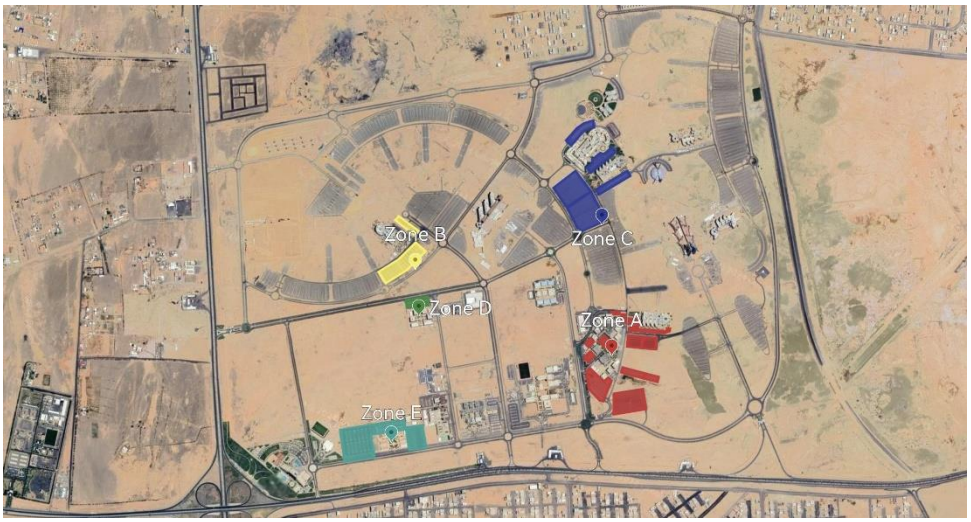
Zone A Includes: Engineering + Pharmacy + Science (Female) + IT (Male) + Buisness (Male)+ Building 13

Zone B includes: Business (Female) + Design (Female)

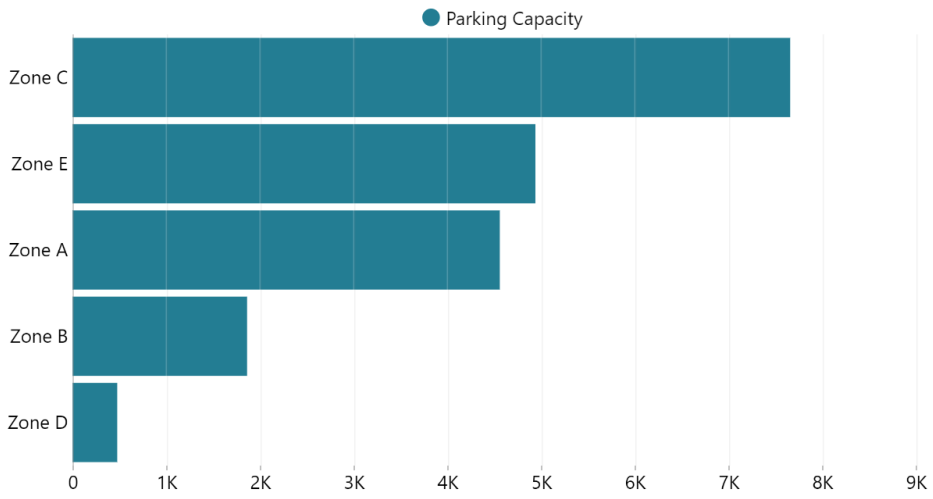
Zone C Includes: Medicine (Male, Female) + College of Applied Science (Male, Female) + Health Science (Female, male) + Nursing (Male and Female)

Zone D Includes: College of Education and Arts (Female)

Zone E Includes: Sharia and Law (Male) + College of Education and Arts (Male)

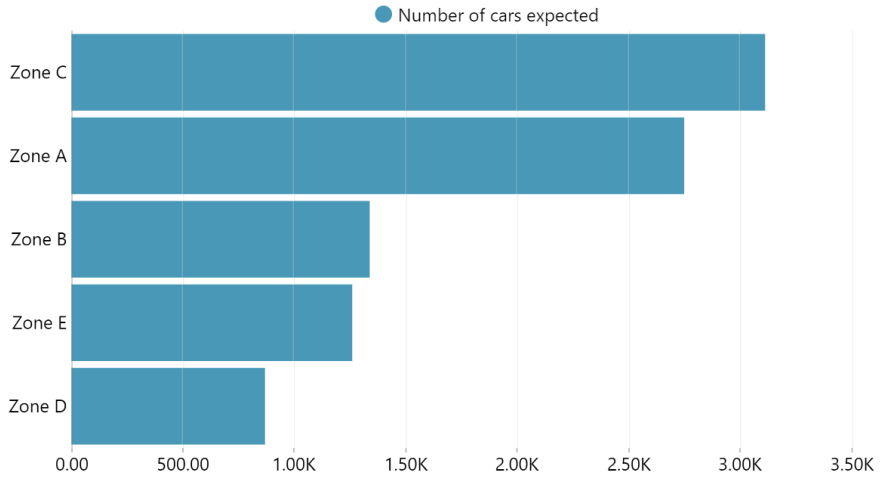


Figures 3. University map.



Figures 4. Parking Capacity.

In calculating the percentage of students coming to the university in their own cars, we relied on [28] survey that the number of students who drive their own cars is 88.5% of the total number of students who come by car, including those who arrive with their parents or friends.



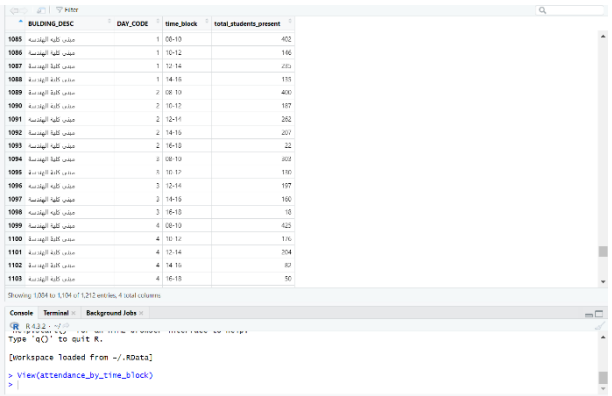
Figures 5. Number of cars expected.

As mentioned in our objectives, we aim to establish parking areas that reflect the university's reputation in terms of aesthetics, design, and alignment with sustainability goals. This involves curbing, planning, and landscaping the parking spaces. Before analyzing the data, we considered these aspects. For example, after determining the total area of the parking spaces in Zone A, we accounted for their design and landscaping before calculating the number of parking spots.

4. Data Analysis and Results

As mentioned in the methodology, we used the R software to analyze and merge the data first. Then, we exported the file to review and modify it in Excel.

```
attendance_by_time_block <- fixing_mistakes_Ver_2 %>%
  mutate(
    START_TIME = format(as.POSIXct(START_TIME, format = "%H:%M:%S"), "%H:%M"),
    START_TIME = as.POSIXct(START_TIME, format = "%H:%M")
  ) %>%
  # Define the time block based on START_TIME
  mutate(
    time_block = case_when(
      START_TIME >= as.POSIXct("08:00", format = "%H:%M") & START_TIME < as.POSIXct("10:00", format = "%H:%M") ~ "08-10",
      START_TIME >= as.POSIXct("10:00", format = "%H:%M") & START_TIME < as.POSIXct("12:00", format = "%H:%M") ~ "10-12",
      START_TIME >= as.POSIXct("12:00", format = "%H:%M") & START_TIME < as.POSIXct("14:00", format = "%H:%M") ~ "12-14",
      START_TIME >= as.POSIXct("14:00", format = "%H:%M") & START_TIME < as.POSIXct("16:00", format = "%H:%M") ~ "14-16",
      START_TIME >= as.POSIXct("16:00", format = "%H:%M") & START_TIME < as.POSIXct("18:00", format = "%H:%M") ~ "16-18",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(time_block)) %>%  # Exclude rows
  group_by(BULDING_DESC, DAY_CODE, time_block) %>%  # Group by building, day, and time block
  summarize(
    total_students_present = sum(ENROLLED_STUDENTS, na.rm = TRUE),  # Sum students in each time block
    .groups = 'drop'
  ) %>%
  arrange(BULDING_DESC, DAY_CODE, time_block)
```



Figures 6. The code and the results in R.

All parking zones were analyzed in relation to the number of students, and the results were as follows:

College	Day	Time Period	Number of Students	Number of Cars Expected	Parking Capacity	Surplus in Parking
Zone 1	Sunday	8:00-10:00	5744	5083	4559	-524
		10:00-12:00	4007	3546	4559	1013
		12:00-14:00	3801	3364	4559	1195
		14:00-16:00	3519	3114	4559	1445
		16:00-18:00	744	658	4559	3901
	Monday	8:00-10:00	5635	4987	4559	-428
		10:00-12:00	4258	3768	4559	791
		12:00-14:00	3945	3491	4559	1068
		14:00-16:00	3539	3132	4559	1427
		16:00-18:00	567	502	4559	4057
	Tuesday	8:00-10:00	5559	4920	4559	-361
		10:00-12:00	3834	3393	4559	1166
		12:00-14:00	3755	3323	4559	1236
		14:00-16:00	3291	2913	4559	1646
		16:00-18:00	768	680	4559	3879
	Wednesday	8:00-10:00	5445	4819	4559	-260
		10:00-12:00	4084	3614	4559	945
		12:00-14:00	3804	3367	4559	1192
		14:00-16:00	3368	2981	4559	1578
		16:00-18:00	515	456	4559	4103
	Thursday	8:00-10:00	2507	2219	4559	2340
		10:00-12:00	2066	1828	4559	2731
		12:00-14:00	1712	1515	4559	3044
		14:00-16:00	802	710	4559	3849
		16:00-18:00	363	321	4559	4238
Zone 2	Sunday	8:00-10:00	3306	2926	1862	-1064
		10:00-12:00	2064	1827	1862	35
		12:00-14:00	1687	1493	1862	369
		14:00-16:00	2219	1964	1862	-102
		16:00-18:00	0	0	1862	1862
	Monday	8:00-10:00	3487	3086	1862	-1224
		10:00-12:00	1816	1607	1862	255
		12:00-14:00	1686	1492	1862	370
		14:00-16:00	1779	1574	1862	288
		16:00-18:00	20	18	1862	1844
	Tuesday	8:00-10:00	3229	2858	1862	-996
		10:00-12:00	2041	1806	1862	56
		12:00-14:00	1656	1466	1862	396
		14:00-16:00	2147	1900	1862	-38
		16:00-18:00	8	7	1862	1855
	Wednesday	8:00-10:00	3424	3030	1862	-1168
		10:00-12:00	1819	1610	1862	252
		12:00-14:00	1676	1483	1862	379
		14:00-16:00	1702	1506	1862	356
		16:00-18:00	20	18	1862	1844
	Thursday	8:00-10:00	845	748	1862	1114
		10:00-12:00	818	724	1862	1138
		12:00-14:00	265	235	1862	1627
		14:00-16:00	78	69	1862	1793
		16:00-18:00	0	0	1862	1862
Zone 3	Sunday	8:00-10:00	6365	5633	7658	2025
		10:00-12:00	3838	3397	7658	4261
		12:00-14:00	4418	3910	7658	3748
		14:00-16:00	5416	4793	7658	2865
		16:00-18:00	321	284	7658	7374
	Monday	8:00-10:00	5330	4717	7658	2941
		10:00-12:00	3762	3329	7658	4329
		12:00-14:00	4423	3914	7658	3744
		14:00-16:00	5333	4720	7658	2938
		16:00-18:00	108	96	7658	7562
	Tuesday	8:00-10:00	5346	4731	7658	2927
		10:00-12:00	3483	3082	7658	4576
		12:00-14:00	3720	3292	7658	4366
		14:00-16:00	5245	4642	7658	3016
		16:00-18:00	86	76	7658	7582
	Wednesday	8:00-10:00	5588	4945	7658	2713
		10:00-12:00	4000	3540	7658	4118
		12:00-14:00	2558	2264	7658	5394
		14:00-16:00	3197	2829	7658	4829
		16:00-18:00	34	30	7658	7628
	Thursday	8:00-10:00	4900	4337	7658	3322
		10:00-12:00	3003	2658	7658	5000
		12:00-14:00	3614	3198	7658	4460
		14:00-16:00	3799	3362	7658	4296
		16:00-18:00	0	0	7658	7658
Zone 4	Sunday	8:00-10:00	1903	1684	476	-1208
		10:00-12:00	1215	1075	476	-599
		12:00-14:00	1065	943	476	-467
		14:00-16:00	1023	905	476	-429
		16:00-18:00	139	123	476	353
	Monday	8:00-10:00	1836	1625	476	-1149
		10:00-12:00	1314	1163	476	-687
		12:00-14:00	1087	962	476	-486
		14:00-16:00	800	708	476	-232
		16:00-18:00	234	207	476	269
	Tuesday	8:00-10:00	2248	1989	476	-1513
		10:00-12:00	1175	1040	476	-564
		12:00-14:00	1138	1007	476	-531
		14:00-16:00	984	871	476	-395
		16:00-18:00	159	141	476	335
	Wednesday	8:00-10:00	1836	1625	476	-1149
		10:00-12:00	1244	1101	476	-625
		12:00-14:00	1031	912	476	-436
		14:00-16:00	893	790	476	-314
		16:00-18:00	254	225	476	251
	Thursday	8:00-10:00	1098	972	476	-496
		10:00-12:00	728	644	476	-168
		12:00-14:00	693	613	476	-137
		14:00-16:00	385	341	476	135
		16:00-18:00	41	36	476	440
Zone 5	Sunday	8:00-10:00	2390	2115	4939	2824
		10:00-12:00	1934	1712	4939	3227
		12:00-14:00	1723	1525	4939	3414
		14:00-16:00	1666	1474	4939	3465
		16:00-18:00	173	153	4939	4786
	Monday	8:00-10:00	2560	2266	4939	2673
		10:00-12:00	1867	1652	4939	3287
		12:00-14:00	1498	1326	4939	3613
		14:00-16:00	1420	1257	4939	3682
		16:00-18:00	366	324	4939	4615
	Tuesday	8:00-10:00	2354	2083	4939	2856
		10:00-12:00	1784	1579	4939	3360
		12:00-14:00	1965	1739	4939	3200
		14:00-16:00	1769	1566	4939	3373
		16:00-18:00	209	185	4939	4754
	Wednesday	8:00-10:00	2567	2272	4939	2667
		10:00-12:00	1925	1704	4939	3235
		12:00-14:00	1632	1444	4939	3495
		14:00-16:00	1463	1295	4939	3644
		16:00-18:00	360	319	4939	4620
	Thursday	8:00-10:00	1245	1102	4939	3837
		10:00-12:00	958	848	4939	4091
		12:00-14:00	865	766	4939	4173
		14:00-16:00	844	747	4939	4192
		16:00-18:00	41	36	4939	4903

Figures 7. Results of our students parking analysis.

Zone A: Congestion was only observed at 8:00 AM on weekdays. This aligns with our firsthand observations as students in this zone, where heavy traffic leads to chaotic and unorganized parking, creating an unpleasant visual impact.

Other zones: No significant issues were found regarding parking availability.

To investigate further whether the occupancy rate during the 8-10 AM period is significantly higher than that during other periods, a Welch Two-Sample t-test was conducted. This test was selected because the data might not have equal variances across the groups. The analysis compared the occupancy percentage during the 8-10 AM period and the other time periods (10-.12, 12-2, etc.) across all five parking areas.

Hypothesis Testing:

Null Hypothesis (H_0): There is no significant difference in the occupancy rates between the 8-10 AM period and other periods.

Alternative Hypothesis (H_1): The occupancy rate during the 8-10 AM period is significantly higher than the other periods.

```r
group_8_10 <- T_TEST_SDP %>%
  filter(`Time Period` == "8:00-10:00") %>%
  select(College, Day, Precentage)

group_other <- T_TEST_SDP %>%
  filter(`Time Period` != "8:00-10:00") %>%
  select(College, Day, Precentage)

# T-Test 8-10 period vs other periods
T_TEST_SDP <- t.test(group_8_10$Precentage, group_other$Precentage,
                     alternative = "greater", var.equal = FALSE)

# Display the T-Test results
print("Overall T-Test results (8:00-10:00 vs other periods):")
print(T_TEST_SDP)

T_TEST_SDP2$Time_Group <- ifelse(T_TEST_SDP2$`Time Period` == "8:00-10:00", "8:00-10:00", "Other")
ggplot(T_TEST_SDP2, aes(x = Time_Group, y = Precentage, fill = Time_Group)) +
  geom_bar(stat = "summary", fun = "mean", position = "dodge") +
  labs(title = "Average Occupancy Rate: 8:00-10:00 vs Other Periods",
       x = "Time Period",
       y = "Average Occupancy Rate (%)") +
  theme_minimal() +
  scale_fill_manual(values = c("8:00-10:00" = "dodgerblue", "Other" = "orange"))
View(T_TEST_SDP)
```

Figures 8. The code used for the T-TEST.

The results of the Welch Two-Sample t-test are as follows:

t = 3.0452

df = 27.523

p-value = 0.002539

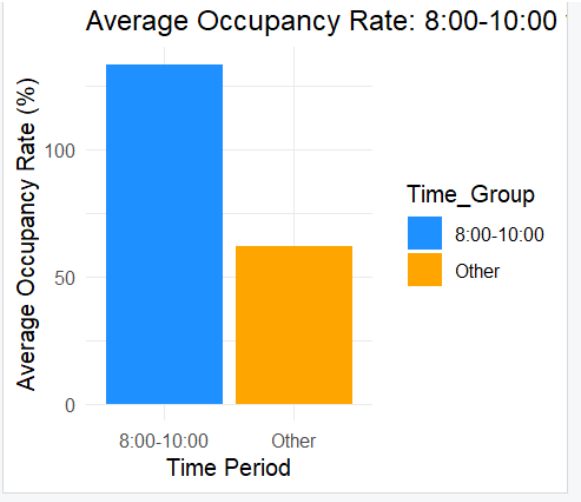
Confidence Interval: [31.64939, Inf] %95

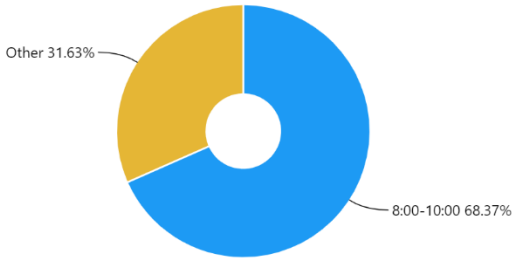
Mean of 8-10 AM period: 133.53%

Mean of other periods: 61.77%

Given the p-value of 0.002539, which is less than the typical significance level of 0.05, we reject the null hypothesis. This indicates that the occupancy rate during the 8-10 AM period is statistically significantly higher than during the other time periods.

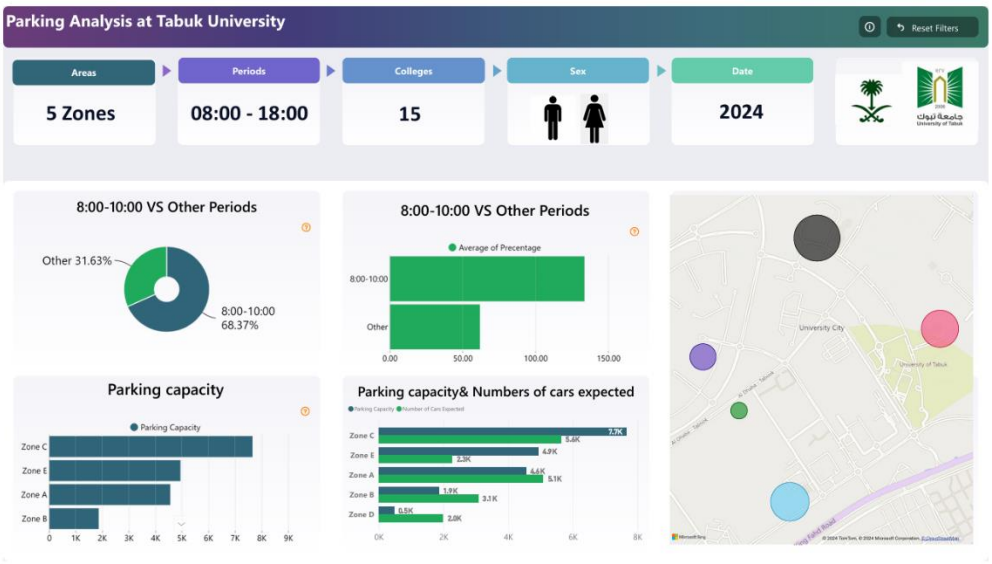
```
data:  group_8_10$Precentage and group_other$Precentage
t = 3.0452, df = 27.523, p-value = 0.002539
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 31.64939      Inf
sample estimates:
mean of x mean of y
133.53471  61.77474

>
```





Figures 9. Results of The T-TEST analysis.

The average occupancy rate during the 8-10 AM period is approximately 133.53%, which is significantly higher than the average occupancy rate during the other periods (approximately 61.77%). The 95% confidence interval indicates that the true difference in means is likely to be at least 31.64%, which further supports the conclusion that the 8-10 AM period represents a critical peak time for parking demand.

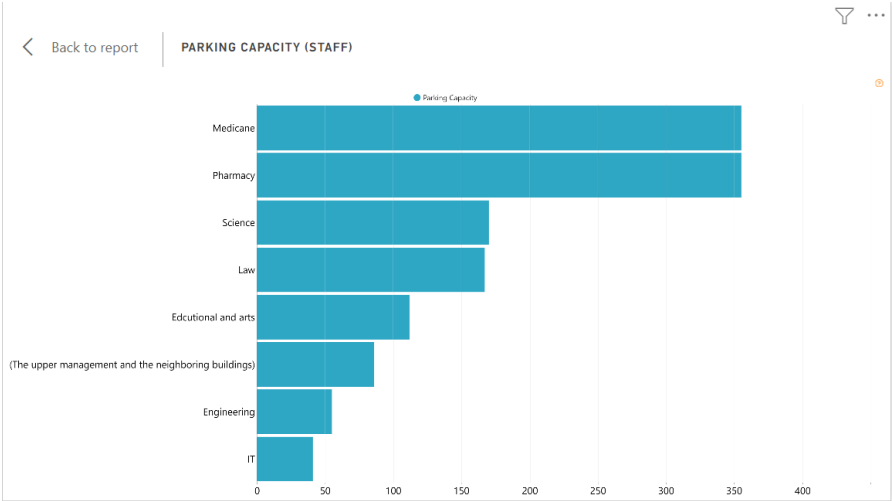


Figures 10. Final analysis for The students parking zones.

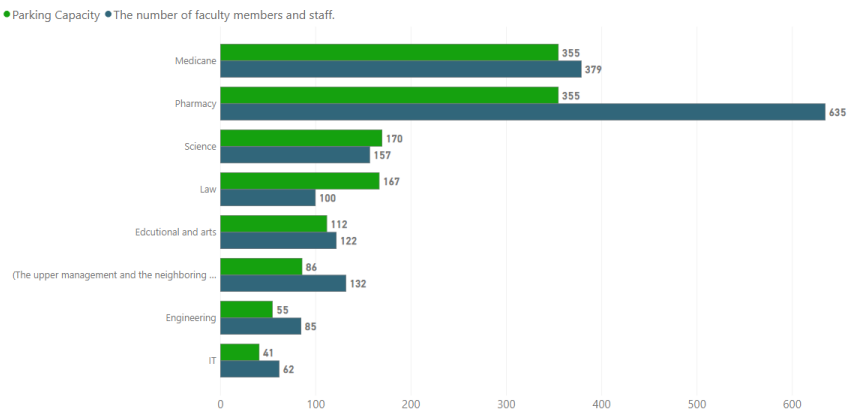
As for faculty and staff parking, the analysis revealed several problems, which were also evident through direct observation.

College	The number of faculty members and staff.	Parking Capacity	Surplus in Parking
IT	62	41	-21
Engineering	85	55	-30
Medicane	379	355	-24
Pharmacy	635	355	-280
Science	157	170	13
Law	100	167	67
Edcutional and arts	122	112	-10
(The upper management and the neighboring buildings)	132	86	-46

Figures 11. Results of our Faculty members and staff parking analysis.



Figures 12. Parking capacity (Staff).



Figures 13. Surplus of parking capacity in the staff parking.

5. Recommendations

After analyzing the data and identifying the underlying problem with parking at the university, which has caused traffic congestion and an unappealing appearance, we propose several solutions to create organized and visually appealing parking areas that reflect the standards of the University of Tabuk.

These suggestions include:

Better Scheduling of Lectures: Adjusting lecture times to move away from the concept that university students must begin their day at 8:00 AM. University students are generally flexible and can attend classes starting later in the morning. This adjustment would help the university organize parking more effectively, as the peak issue lies within the 8:00–10:00 AM time frame.

Implementing Curbs in Parking Areas: Adding curbs in the parking areas to prevent visual clutter caused by unorganized spaces and ensuring proper parking alignment.

Designating Pedestrian Walkways: Creating designated pedestrian pathways for individuals after they exit their vehicles, so they don't have to walk between parked cars, enhancing safety and organization.

Supporting Sustainability: Incorporating greenery by planting trees in parking areas, promoting a more sustainable and pleasant environment.

Introducing Premium Parking Spaces: Allocating certain convenient and easily accessible parking spaces near buildings as premium parking areas. However, access to these spaces would require students to achieve a specific GPA or higher. This system would not only provide economic benefits for students but also serve as a motivational tool to improve academic performance.

Monitor Utilization Regularly: Conduct periodic reviews of staff parking usage to evaluate the need for expansion or redistribution of spaces.

Expand Parking Capacity for Staff: If feasible, create additional parking spaces designated exclusively for staff to address shortages.

6. Conclusion

The project provided an analysis highlighting clear issues for the university regarding lecture scheduling, parking organization, and layout planning, as well as addressing the current visual distortion. Our recommendations emphasized covering all these aspects, which, if implemented, would result in more organized and visually appealing parking spaces. This, in turn, would lead to reduced traffic congestion and offer a significant partial solution to the broader traffic problem.

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